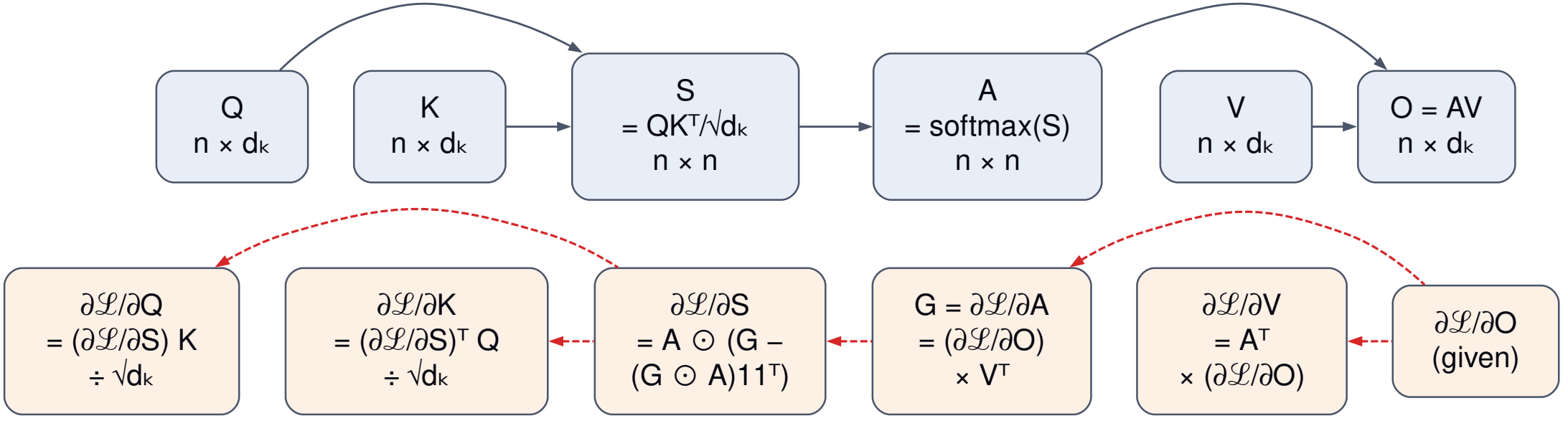
Q
n × dk
K
n × dk
S
= QKT/√dk
n × n
A
= softmax(S)
n × n
V
n × dk
O = AV
n × dk
∂ℒ/∂Q
= (∂ℒ/∂S) K
÷ √dk
∂ℒ/∂K
= (∂ℒ/∂S)T Q
÷ √dk
∂ℒ/∂S
= A ⊙ (G –
(G ⊙ A)11T)
G = ∂ℒ/∂A
= (∂ℒ/∂O)
× VT
∂ℒ/∂V
= AT
× (∂ℒ/∂O)
∂ℒ/∂O
(given)